NORTHWESTERN POLYTECHNIC

DEPARTMENT OF SCIENCE

COURSE OUTLINE – Fall 2026

EG1300 (A2): Engineering Mechanics – 4 (3-0-2) 75 Hours for 15 Weeks

Northwestern Polytechnic acknowledges that our campuses are located on Treaty 8 territory, the ancestral and present-day home to many diverse First Nations, Metis, and Inuit people. We are grateful to work, live and learn on the traditional territory of Duncan's First Nation, Horse Lake First Nation and Sturgeon Lake Cree Nation, who are the original caretakers of this land.

We acknowledge the history of this land and we are thankful for the opportunity to walk together in friendship, where we will encourage and promote positive change for present and future generations.

INSTRUCTOR: Dr. Braden Kelly
OFFICE: J218
OFFICE HOURS: Unrestricted drop-in/by-appointsment

PHONE: (780) 539-2963
E-MAIL: bkelly@nwpolytech.ca

CALENDAR DESCRIPTION:

Equilibrium of planar systems. Analysis of statically determinate trusses and frames. Friction. Centroids and centres of gravity. Forces and moments in beams. Second moments of area. **Restricted to Engineering students.**

COREQUISITE(S):

MA1000

REQUIRED MATERIALS:

Hibbeler, R. C. - *Engineering Mechanics: Statics*. 15th Edition. Prentice Hall/Pearson.

RECOMENDED TEXT/RESOURCE MATERIALS:

Engineering Mechanics, Statics, University of Alberta's open educational resources. Free open-access via https://engcourses-uofa.ca/books/statics/

Copyright © 2024, Northwestern Polytechnic and its licensors.

NORTHWESTERN POLYTECHNIC

DELIVERY MODE(S):

Lecture Tuesday & Thursday 10:00 – 11:20am, room D308.
Lab L1: Wednesday 2:30-4:20pm, room J204.
Lab L2: Friday 2:30-4:20pm, room B305.
Lab L3: Monday 2:30-4:20pm, room B201.

LEARNING OUTCOMES:

Students will have the knowledge to be able to analyze and calculate the forces and moments acting on members of 2-D and 3-D planar structures in equilibrium such as trusses and frames. They will be able to include various 1-D and 2-D distributed loading configurations using centroids and centers of gravity. (the following list is taken from the course outline at UofA for ENGG 130:

1. Conduct basic vector operations for solving engineering mechanics problems such as dot product and cross product using force and position vectors etc.
2. Draw free body diagrams for 2D and 3D particles and rigid bodies
3. Write & solve force equilibrium equations for particles in 2D and 3D
4. Calculate the moment of forces in 2D and 3D for solving engineering mechanics problems
5. Write and solve the force and moment equilibrium equations for rigid bodies in 2D and 3D.
6. Analyze simple structures such as trusses, frames and machines for calculating the internal forces and internal moments.
7. Employ the concept of static equilibrium for evaluation of the friction problems
8. Calculate centroids, centre of mass, and moments of inertia for application in EG1300 and in subsequent courses.

TRANSFERABILITY:

Please consult the Alberta Transfer Guide for more information. You may check the transferability of this course at the Alberta Transfer Guide main page http://www.transferalberta.alberta.ca.

** For courses with alpha (letter) grading, a grade of D or D+ may not be acceptable for transfer to other post-secondary institutions. **Students are cautioned that it is their responsibility to contact the receiving institutions to ensure transferability.**

Copyright © 2024, Northwestern Polytechnic and its licensors.

NORTHWESTERN POLYTECHNIC

EVALUATIONS:

Component	Weight	Comment
Assignments	10%	Weekly (Done on Eng. Paper with Cover Page)
Lab assignments	20%	Assigned problems are due at end of each lab.
Midterm Exam	30%	TBD (Mid October)
Final Exam	40%	TBA, Cumulative exam.
Total	100%	

Note: Assignment weight may be downgraded to 5%, in which case Final Exam will carry a weight of 45%.

GRADING CRITERIA:

Please note that most institutions will not accept your course for transfer credit **IF** your grade is **less than C-**.

Alpha Grade	4-point Equivalent	Percentage Guidelines	Alpha Grade	4-point Equivalent	Percentage Guidelines
A+	4.0	95-100	C+	2.3	67-69
A	4.0	85-94	C	2.0	63-66
A-	3.7	80-84	C-	1.7	60-62
B+	3.3	77-79	D+	1.3	55-59
B	3.0	73-76	D	1.0	50-54
B-	2.7	70-72	F	0.0	00-49

Copyright © 2024, Northwestern Polytechnic and its licensors.

NORTHWESTERN POLYTECHNIC

COURSE SCHEDULE/TENTATIVE TIMELINE:

Week	Date	Topics	UofA Textbook	Hibbeler	Note
1	Sep. 03	Intro to EG1300 Scalars & Vector (Lec. 1)	2.2 2.4	Chapter 1,2	
2	Sep. 8/10	Cartesian vector notation, addition (Lec. 2) Position vectors (Lec. 3) Force directed along a line (Lec. 3)	2.6 3.1 2.7	Chapter 2	HOA #1 due Lab #1
3	Sep. 15/17	Dot product (lec. 4) Particle equilibrium (2D) (Lec. 5) Free body diagrams (Lec. 5)	5.1 4.4	Chapter 3	HOA #2 due Lab #2
4	Sep. 22/24	co-planar and 3D force systems (Lec. 6) Vector cross product (Lec 7) moment of a force (Lec 7)	2.8 3.2	Chapter 4	HOA #3 Lab #3
5	Sep. 29/Oct. 01	Moment about an axis (Lec. 8) Moment of a couple Reduction of force and couple system	3.2.3 3.3 3.4	Chapter 4	HOA #4 due Lab #4
6	Oct. 06/8	Reduction of simple distributed loads (Lec 10) Distributed Loads (Lec. 11)	3.5	Chapter 4/5	HOA #5 Lab #5
7	Oct. 13/15	Support Reactions (Lec. 12) Two and three-force members (Lec. 12)	5.2	Chapter 5	HOA #6 Lab #6
		MIDTERM			
8	Oct. 20/22	Plane trusses (Lec. 13) Method of joints (Lec. 13) Method of sections (Lec. 14) Zero Force Members (Lec. 14)	6.1	Chapter 6	HOA #7 due Lab #7
9	Oct. 27/29	Internal forces in members (Lec. 15) Shear force and bending moments (Lec. 16)	7.1-7.2	Chapter 7	HOA #8 Lab #8
10	Nov. 03/05	Relationships of distributed load, SF, & BM Graphical approach for diagrams (Lec. 17) Examples (Lec. 18)	7.3 7.2	Chapter 7	HOA #9 Lab #9
11	Nov. 10 - 14	Reading Week			
12	Nov. 17/19	Centre of gravity, centroids (Lec. 19) centroids by integration and composite bodies (Lec. 20)	9.1 - 9.3	Chapter 9	HOA #10 due Lab #10
13	Nov. 24/26	Moment of inertia for areas (Lec. 21) Moment of inertia by integration (Lec. 22) Parallel axis theorem (Lec. 22)	10.1 10.3 - 10.4	Chapter 10	HOA #11 due Lab #11
14	Dec. 01/03	Dry friction (Lec. 23 & 24)	8.1 - 8.2	Chapter 8	HOA #12 Lab #12
15	Dec. 09	course review			HOA #13

Copyright © 2024, Northwestern Polytechnic and its licensors.

NORTHWESTERN POLYTECHNIC

STUDENT RESPONSIBILITIES:

CALCULATOR POLICY: Any calculator without communications features that is approved by UAlberta Engineering faculty (*e.g.* TI-36XPro / TI-30XII) may be used during EG1300 examinations. Smartphones, smart glasses, smart watches, Tablets/Laptop computers *etc.* are prohibited. Cellular phones must be shut off during exams. All calculators with removable covers must have the covers removed and stored elsewhere during the exam.

LABORATORIES:

- The first lab starts in the second week of the semester
- Students will complete engineering calculations during each lab session. Bring the required tools e.g., pencil, calculator, ruler(straight edge), engineering paper, stapler, etc.
- No smart or electronic devices are allowed whatsoever. Only printed notes, textbook and printed formula sheet may be brought in.
- Labs must be completed on Engineering Paper and submitted, stapled, with cover page along with the question sheet, at the end of the lab session.
- Use appropriate significant figures. Circle and label your answers.
- All work submitted must be your own.
- The final grade will be on the best 10 out of 12 labs. Labs marked 0 will not be eligible for removal from the best 10 of 12.
- Marks will be deducted for messy work, no title page, failing to circle and label answers, submitting on non-engineering paper or failing to staple pages together neatly. It is entirely possible to get a mark below 20% and have every single final answer correct. Marks are for correct and clear procedure not final answers!

STATEMENT ON ACADEMIC MISCONDUCT:

Academic Misconduct will not be tolerated. For a more precise definition of academic misconduct and its consequences, refer to the Student Rights and Responsibilities policy available https://www.nwpolytech.ca/about/polytechnic-leadership/policies-directory.

**Note: all Academic and Administrative policies are available on the same page.

Copyright © 2024, Northwestern Polytechnic and its licensors.

NORTHWESTERN POLYTECHNIC

Additional Information:

- Late work receives a grade of zero. 1 minute past due date is considered late.
- Make-up assignments are not allowed for any assessment.
- Students are expected to come to lectures having looked at the relevant section of the textbook beforehand.
- Assessments that are not done on Engineering paper and to the format specified will receive a grade of zero.
- Engineering is about showing your work carefully and clearly. Final answers alone are not worth many points.
- During Midterms and Final exams students must remain seated. There are no washroom breaks.

Additional Additional Information:

Engineers Rule The World (ERTW)

Copyright © 2024, Northwestern Polytechnic and its licensors.